

1. Describe the cross-drainage structures and sub-surface drainage.

Answer:

Cross-drainage structures are those structures provided where a road, railway, or canal crosses a natural stream, river, or drainage channel. Their main purpose is to allow water to pass safely across the route without damaging the road or interrupting the natural flow.

Types of cross drainage structures:

i. Culvert:

- Used when the waterway is upto 6 m.
- It is a closed conduit placed under the embankment to carry water cross the roadway.
- Culverts are preferred than minor bridges since the culvert is always beneath the carriageway whereas bridge surface forms the part of carriageway.



Types of culverts:

- **Box culvert:** These culverts are constructed where the nature of the soil below the foundation is not suitable for the individual footing under piers and abutments.
- **Pipe culvert:** Used when the stream carries low discharge and alignment has high embankment.
- **Slab culvert:** These culverts have masonry abutment with stone slab over them and mostly used where stone is easily available in the locality.
- **Arch culvert:** This type of culvert is preferred under the conditions where high fillings are involved and there are heavier loadings on the culvert.

ii. Bridge:

- Bridge is constructed when the linear waterway is more than 6m.
- It is a structure constructed over the water course to carry traffic over it.
- Bridges may be classified either from the materials used to construct superstructure such as steel bridges, concrete bridges, timber bridges etc.
- From the structural point of view, it can be divided as suspension, movable etc.

- According to the span length it can be divided as minor bridge, major bridge or long bridge.
- **Parts of bridge:**
 - a. Foundation:
 - b. Substructure:
 - c. Superstructure:



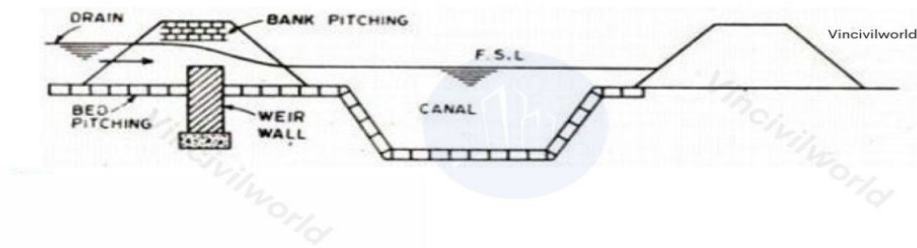
iii. Causeway:

- It is provided instead of culverts on less important road which saves the construction cost.
- These are provided where the maximum depth of flow does not exceed 1.5m.
- Causeways may be high level or low level.
- Low level causeway (Irish bridge) is at the bed level of the stream which remains dry for most
- High level causeway (submersible bridge) is quite above the stream bed and is provided to allow normal floods to pass through them.



iv. Aqueduct:

- Aqueduct is an open or closed conduit sufficiently above the roadway to drain water across the road.
- Such overhead structures with the provision of pillar supports on either side of the road can be advantageously used in hill roads where culverts are not feasible.



Subsurface drainage:

Drainage can be defined as the process of interception and removal of water from over, under and vicinity of road surface. It can be surface drainage or subsurface drainage. Removal and diversion of surface water from highway and adjoining land is called surface drainage. Similarly, removal of water from below the ground surface is called subsurface drainage.

Objectives of subsurface drainage:

- i. To lower the groundwater table.
- ii. To prevent water accumulation beneath pavements.
- iii. To improve soil strength and bearing capacity.
- iv. To reduce pavement damage caused by water.
- v. To prevent frost action and soil instability.

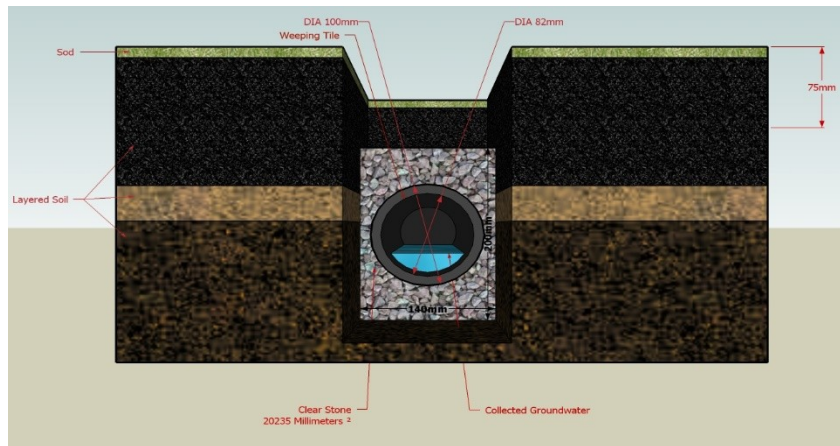
Components of subsurface drainage:



- i. **Perforated drain pipe:** Collects groundwater.
- ii. **Filter material (gravel/sand):** Prevents soil from entering the pipe.
- iii. **Geotextile filter:** Stops fine particles from clogging the drain.
- iv. **Collector drain:** Carries collected water to the outlet.
- v. **Outlet:** Discharges water into a natural drain or channel.

Types of subsurface drainage:

- i. **Longitudinal drains:** Installed parallel to the road.
- ii. **Transverse drains:** Installed across the road to intercept groundwater.
- iii. **French drains:** Gravel-filled trenches with perforated pipes.



- iv. **Interceptor drains:** Constructed uphill to intercept seepage before it reaches the road.

Advantages:

- i. Increases pavement life.
- ii. Prevents weakening of subgrade soil.
- iii. Reduces maintenance costs.
- iv. Improves road safety and durability.
- v. Controls seepage and waterlogging.

Cross-drainage structures ensure the safe passage of surface water across roads, railways, or canals, while sub-surface drainage systems remove excess groundwater from beneath the ground. Both systems are essential for maintaining the stability, durability, and safety of transportation infrastructure.

2. How many cables are there in suspension bridge? Describe the function of each type. How is the load on the bridge transferred to the ground?

Answer:

A suspension bridge primarily uses three types of cables to support and stabilize the bridge. Each cable has a specific function in transferring loads safely to the towers and anchorages.

i. Main cable (Suspension Cable):

- It is the primary load-carrying cable of the bridge.
- It passes over the tops of the towers and is anchored firmly at both ends.
- Usually made of thousands of high-strength steel wires bundled together.
- Carries the load of the bridge deck and traffic through the suspenders.
- Transfers the load to the towers and anchor blocks.

Function:

- Supports the entire bridge deck.
- Transfers loads to towers and anchorages through tension.

ii. Hanger cables (Suspenders):

- These are **vertical steel cables or rods** connecting the bridge deck to the main cable.
- Placed at regular intervals along the span.

Function:

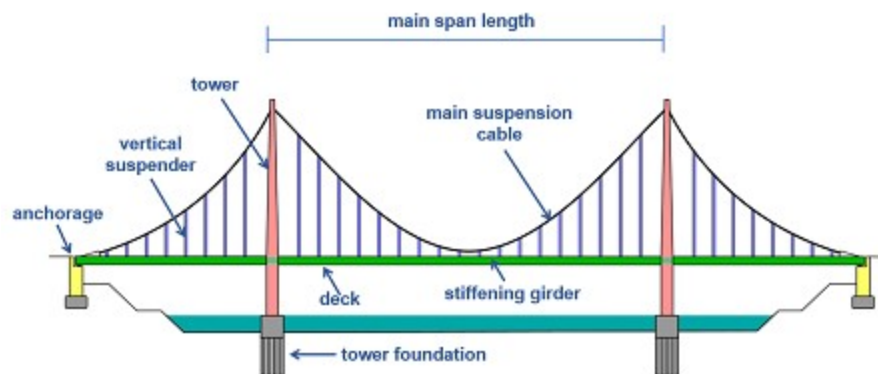
- Transfer the weight of the bridge deck and moving vehicles to the main cable.
- Keep the bridge deck suspended and properly aligned.

iii. Anchor cables:

- These are the portions of the main cable extending from the towers to the anchor blocks.
- They are firmly fixed into massive concrete anchorages.

Function:

- Balance the tension in the main cable.
- Transfer tensile forces safely to the ground through anchor blocks.
- Prevent the bridge from collapsing toward the center span.



In a suspension bridge, the load from vehicles and the bridge itself is transferred to the ground through a sequence of structural members.

Load Transfer Mechanism in flowchart diagram:

Traffic Load + Self-weight of Deck



Bridge Deck



Hanger Cables (Suspenders)



Main Cables



Towers (Pylons)



Anchor Blocks



Foundations



Ground (Soil/Rock)

- **Bridge deck:**
It carries traffic loads and transfers them to the hanger cables.
- **Hanger cables:**
Hanger cables also known as suspenders carry tension.
- **Main cables:**
Main cables also called suspension cables also carry tension.
- **Towers:**

- They carry compression.
- **Anchor blocks:**
These blocks resist tension and prevent the main cables from pulling inward.
- **Foundations:**
Foundation of the bridge finally distributes all forces safely to the supporting soil or rock.

3. In any transportation planning exercise, what are the major technical considerations to be given attention to. In your objective view, why very often our planning exercise fail to address the main issues?

Answer:

Transportation planning should ensure that the system is safe, efficient, economical, environmentally sustainable, and capable of meeting future travel demand. The major technical considerations include:

- i. **Traffic Demand Analysis**
 - a. Present and future traffic volumes are studied.
 - b. Travel demand based on population and economic growth are forecasted.
- ii. **Route Selection and Alignment**
 - a. The most economical and technically feasible route is selected.
 - b. Travel time, construction cost, and environmental impacts are minimized.
- iii. **Land Use Planning**
 - a. The transport facilities with present and future land-use patterns are coordinated.
 - b. Balanced urban and regional development is encouraged.
- iv. **Engineering Design Standards**
 - a. An appropriate geometric design, pavement design, bridges, drainage, and safety standards are adopted.
- v. **Environmental and Social Impact Assessment**
 - a. Pollution, noise, deforestation, displacement, and ecological damage must be minimized.
 - b. Compliance must be ensured with environmental regulations.
- vi. **Economic and Financial Analysis**
 - a. Conduct cost–benefit analysis is conducted.
 - b. Project affordability, funding sources, and long-term maintenance costs are assessed.
- vii. **Road Safety**
 - a. Adequate sight distance, road signs, markings, lighting, pedestrian facilities, and traffic control devices are provided.
- viii. **Drainage and Climate Resilience**
 - a. Proper surface and subsurface drainage is designed.
 - b. Floods, landslides, earthquakes, and climate change impacts must be considered.
- ix. **Public Transport and Accessibility**
 - a. Roads must be integrated with public transport systems.
 - b. Accessibility must be ensured for pedestrians, cyclists, elderly people, and persons with disabilities.
- x. **Operation and Maintenance**
 - a. Plans for regular maintenance and efficient traffic management to ensure long service life are ensured.

In many developing countries, including Nepal, transportation planning often fails due to the following reasons:

- **Inadequate and unreliable** traffic and socioeconomic **data** leading to poor forecasting.
- **Political interference** in project selection and prioritization instead of technical evaluation.
- **Weak coordination** among government agencies and stakeholders.

- **Insufficient public participation**, resulting in plans that do not reflect local needs.
- **Limited financial and technical resources** for proper studies and implementation.
- **Poor enforcement** of planning regulations and development controls.
- **Rapid and unplanned urbanization**, making plans obsolete before implementation.
- **Neglect of environmental and social impacts**, causing delays and public opposition.
- **Poor maintenance planning**, resulting in rapid deterioration of infrastructure.
- **Weak institutional capacity** and shortage of skilled professionals.

4. In context of scarcity of landfill sites, solid waste management has become difficult task in Kathmandu Valley. What are your suggestions to reduce and recycle the wastes so that pressure on landfill site could be out down? Provide pragmatic solutions.

Answer:

In Nepal, one of the most talked about landfill site is **Banchare Danda landfill site**, serving as the permanent disposal site for the Kathmandu Valley receiving about 1500 metric tons of daily wastages. Locals of the area frequently block garbage trucks due to unfulfilled government agreements, foul odors, and severe environmental/health hazards. Another landfill site located nearby Kathmandu is **Sisdol landfill site** which was intended to be a 3-year temporary dump, but operated for almost two decades. Enormous quantity of wastage accumulation on the area resulted in groundwater contamination and disease. The situation led to years of protests before the shift to Banchare Danda.

Some suggestions to reduce and recycle the wastes so that pressure on landfill site could be out down are listed and described below:

i. Composting:

- Composting is the aerobic biological decomposition of biodegradable organic matter in which putrescible organic matters are decomposed into stable end products, and converted into compost.
- Compost has high soil conditioning value which is suitable for agriculture and horticulture.
- The method helps in reducing disease vectors, produce safer water disposal and improves agricultural productivity.



- There are three methods of composting:
 - a. Composting by trenching:**
 - Trench are excavated and refuse is laid in the layers of 15 cm in the trench.
 - Bacterial decomposition of organic matter take place which increases the temperature up to 75 degrees.
 - After 4 to 6 months, solid waste completely stabilizes which can be used as manure.
 - b. Open windrow composting:**
 - Refuse is laid in the form of piles or windrow where the bacterial decomposition of organic matter takes place.
 - The waste gets completely stabilized in 7 to 10 weeks.
 - c. Mechanical composting:**
 - Piles of solid waste is turned by mechanical device.

- The stabilization of waste takes place in 1 to 2 weeks and can be used as compost.

ii. Vermicomposting:

- It is the method of solid waste management where the organic waste is decomposed using earthworms.
- It produces high-quality compost, faster than conventional composting and is eco-friendly.
- However, the method is sensitive to temperature and moisture, suitable only for biodegradable waste.
- The method improves soil fertility and encourages organic farming.

iii. Recycling:

- Materials are collected, sorted, processed and manufactured into new products.
- Recyclable materials include: paper, glass, plastics, aluminum, steel etc.
- It conserved natural resources, save energy and reduces landfill demand.
- However, recycling process requires source segregation and the contamination might lower the recycling efficiency.
- It helps in cleaner surroundings and reduced accumulation of wastes.

iv. Adoption of the 3R Principle

The most effective way to lower landfill pressure is to integrate the **Reduce, Reuse, and Recycle (3R)** concepts at the source.

- **Source Segregation:** Mandatory separation of waste by households and enterprises into organic, recyclable, and hazardous categories is essential for the success of any recycling or composting program.
- **Reuse of Materials:** Encouraging the reuse of items such as plastic containers for household storage, seedling nurseries, or home-made drip irrigation systems extends the lifecycle of non-biodegradable products.
- **Material Recycling:** Formal systems should be established to collect and process paper, glass, plastics, and metals. This not only saves energy and natural resources but also directly reduces the demand for landfill space.

Hence, the practice of waste separation from the generators helps in proper waste management by reducing the pressure on landfill sites. Also, the quantity of wastages generated from each household and enterprises must be reduced by adopting the concept of recycle and reuse. Different organizations such as Doko Recyclers, smartpaani, SAHAS Nepal are also working actively in grassroot level to encourage people for adopting alternative water recycling practices such as greywater management, eco-friendly waste collection etc.

5. You are required to supply drinking water to some community in a hilly region. Describe the various steps you will take to achieve this in the context of Nepal.

Answer:

In a hilly region, adequate head can be achieved so that we can design gravity water supply project. The steps that can be involved in supplying drinking water to community in a hilly region are stated below:

i. Data collection and selection of design period:

- At first, survey is carried out to determine the discharge available in the source, population demand, topography, route length, and levels of intake, pipe alignment, and reservoir.
- Appropriate design period for the project must be selected to estimate the future population by the help of population forecasting methods.

ii. Estimation of water demand:

- Per capita water demand is calculated.
- Determine the total design discharge:

$$Q = P \times q$$

Where:

Q = Total water demand (L/day)

P = Design population

q = Per capita demand (L/capita/day)

iii. Selection and layout of route:

Topographic and engineering surveys are carried out to determine the pipe alignment, ground profile, and elevations of source and destination.

iv. Calculation of available head:

Determine the difference in elevation between the water source and the receiving reservoir.

$$H = Z_1 - Z_2$$

Where:

H = Available head (m)

Z_1 = Elevation of source water level

Z_2 = Elevation of receiving reservoir water level

v. Fix the design flow:

The maximum daily or peak hourly flow that the transmission main should carry is estimated.

vi. Selection of pipe material:

Suitable pipe material for example: DI, MS, HDPE, PVC, etc. based on pressure, durability, cost, and site conditions are selected.

vii. Determination of pipe diameter:

A suitable flow velocity, usually 0.6–3.0 m/s is assumed.

Compute pipe diameter using:

$$Q = \frac{\pi D^2}{4} V$$

Where:

Q = Discharge (m³/s)

D = Diameter (m)

V = Velocity (m/s)

viii. Head loss calculation:

Calculate friction losses using the Hazen–William's equation:

$$h_f = \frac{10.67 L Q^{1.852}}{C^{1.852} D^{4.87}}$$

Where:

h_f = Friction head loss (m)

L = Length of pipe (m)

Q = Discharge (m³/s)

D = Diameter (m)

C = Hazen-William's coefficient

Also, minor losses due to bends, valves, entrances, and exits are also considered.

ix. Check HGL and pressure conditions:

Hydraulic Adequacy is checked by:

$$H \geq h_f + h_m$$

Where:

H = Available head

h_f = Friction loss

h_m = Minor losses

If not satisfied, pipe diameter is increased and calculations are repeated. The positive pressure must be ensured throughout the pipeline. Excessive pressures that may damage the pipe must be avoided.

x. Design of appurtenances:

Appurtenances such as valves, pumps, bends, fittings must be appropriately designed. Provide air valves at high points, washout valves at low points, isolation valves at suitable intervals.

6. What is the difference between the renewable and non-renewable energy? Also write down the advantages and disadvantages of solar energy, bio-gas and hydropower.

Answer:

Renewable energy is clean, sustainable, and environmentally friendly, making it the preferred option for future energy development. Non-renewable energy provides reliable power but is finite, polluting, and unsustainable. The differences between renewable and non-renewable energy are mentioned below:

Renewable energy	Non-renewable energy
i. Energy obtained from the natural sources that are replenished continuously is known as renewable energy.	i. Energy obtained from finite natural resources that cannot be replenished within a human lifetime is known as non-renewable energy.
ii. It is sustainable source of energy and can be used repeatedly.	ii. It is exhaustible source of energy and will eventually run out.
iii. It produces little to no pollution and greenhouse gasses.	iii. it produces significant pollution and greenhouse gas emissions.
iv. It is environmentally friendly.	iv. it causes environmental degradation and contributes to climate change.
v. Renewable energy has high installation cost but low operating cost.	v. Non-renewable energy has lower initial cost but higher fuel and operating costs over time.
vi. It is suitable for long-term energy security.	vi. it is not sustainable for long-term energy needs.
vii. Example: solar, wind, hydropower	vii. Example: coal, petroleum, nuclear fuel etc.

i. Solar energy:

Solar energy provides about 7% of the world's electricity supply but only 2 to 3% of total global primary energy consumption.

Advantages

- Renewable and inexhaustible source of energy.
- Environment-friendly with no air pollution during operation.
- Suitable for remote and rural areas where grid electricity is unavailable.
- Low operation and maintenance costs.
- Reduces dependence on fossil fuels.

Disadvantages

- High initial installation cost.
- Electricity generation depends on sunlight and weather conditions.
- Requires large land area for large-scale solar plants.
- Energy storage (batteries) increases system cost.
- Lower efficiency during cloudy or rainy seasons.

ii. Biogas:

Biogas fulfills less than 1% of the total global energy demand.

Advantages

- Renewable source produced from organic wastes such as animal dung and kitchen waste.
- Provides clean fuel for cooking and lighting.
- Reduces indoor air pollution and firewood consumption.
- Produces nutrient-rich slurry that can be used as organic fertilizer.
- Helps in proper waste management and sanitation.

Disadvantages

- Requires a regular supply of organic waste and water.
- Gas production decreases during cold weather.
- High initial construction cost for the digester.
- Requires proper maintenance and technical knowledge.
- Not suitable where sufficient organic waste is unavailable.

iii. Hydropower:

Hydropower fulfills 14-15% of the global electricity demand.

Advantages

- Renewable and clean source of energy.
- Low operating and maintenance costs after construction.
- High efficiency and long service life.
- Produces very low greenhouse gas emissions.
- Nepal has abundant hydropower potential due to its rivers and steep terrain.

Disadvantages

- Very high initial construction cost.
- Construction period is long.
- Can affect aquatic ecosystems and local communities.
- Power generation depends on river flow, which may reduce during the dry season.
- Large projects may require relocation of people and environmental mitigation measures.

7. If you were to design an irrigation system to be totally maintained and managed by local farmers after construction. What factors would you consider in deciding about line and unlined sections of canal? Give technical and economic reasons and discuss.

Answer:

The choice between **lined** and **unlined canals** depends on technical, economic, environmental, and social factors. Since the irrigation system will be operated and maintained by local farmers, the design should be simple, affordable, durable, and easy to maintain while ensuring efficient water delivery. This kind of irrigation system is widely known as Farmers Managed Irrigation System (FMIS). In Nepal, **Farmer-Managed Irrigation Systems** cover about 70% of the country's irrigated agricultural land. Some of the biggest FMIS maintained in Nepal are Rajapur Irrigation System, Rani Jamara Kulariya Irrigation System etc.

The factors that must be considered on deciding about line and unlined sections of canal are listed below:

a. Technical factors:

i. Soil Type

- **Permeable soils (sand, gravel):** Lined canals are preferred to reduce seepage losses.
- **Impermeable soils (clay):** Unlined canals may be adequate due to low seepage.

ii. **Seepage Losses**

- High seepage areas require lining to conserve water.
- Where water is abundant and seepage is low, unlined canals can be adopted.

iii. **Topography**

- In steep or hilly terrain, lining helps prevent erosion and slope failure.
- In flat terrain with stable soil, unlined canals may perform satisfactorily.

iv. **Water Availability**

- If water is scarce, lined canals improve conveyance efficiency.
- If water is plentiful, limited seepage losses may be acceptable.

v. **Maintenance Requirements**

- Lined canals require less routine maintenance.
- Unlined canals need frequent desilting, weed removal, and bank repairs.

vi. **Sedimentation and Erosion**

- Lining reduces erosion and minimizes sediment deposition.
- Unlined canals are more susceptible to erosion and silting.

vii. **Groundwater Conditions**

- In waterlogged areas, lining reduces seepage and helps prevent further waterlogging.
- In groundwater recharge areas, controlled seepage from unlined canals may be beneficial.

b. Economic factors:

i. **Initial Construction Cost**

- Lined canals have higher construction costs.
- Unlined canals are cheaper to construct.

ii. **Operation and Maintenance Cost**

- Lined canals have lower long-term maintenance costs.
- Unlined canals require regular maintenance, increasing long-term expenses.

iii. **Water Saving**

- Lined canals reduce water losses, increasing irrigation efficiency and crop productivity.
- Water savings can justify the higher initial investment where water is limited.

iv. **Availability of Local Materials**

- If cement, concrete, or masonry materials are expensive or difficult to transport, unlined canals may be more practical.
- Where local stone and skilled labor are available, lining becomes more economical.

v. **Availability of Local Labor**

- Farmer-managed systems can maintain unlined canals using local labor.
- Lined canals may require skilled workers for repairs.

A combination of lined and unlined canals is generally the most practical and economical solution.

8. **Design an irrigation canal of 15 m³/s capacity, whose side slope coefficient and bed slope are 0.65 and 0.02% respectively with Manning's roughness coefficient of 0.023 so that its flow velocity could be laid in between non-silting and non-scouring value.**

Answer:

∴ design the irrigation canal, we use Kennedy's theory together with Manning's equation

$$\text{Discharge, } Q = 15 \text{ m}^3/\text{s}$$

$$\text{Sideslope coefficient, } z = 0.65 (H : V)$$

$$\text{Bed slope, } S = 0.02\% = 0.0002$$

$$\text{Manning's roughness coefficient, } n = 0.023$$

The canal should flow at a velocity between the non-silting $\wedge$ non-scouring limits.

Kennedy's critical velocity equation:

$$V = 0.55 m,$$

$$V = 0.55 m D^{0.64}$$

Assuming critical velocity ratio $m = 1.0$,

$$V = 0.55 D^{0.64} \quad V = 0.55 D^{0.64} \quad V = 0.55 D^{0.64}$$

$$\text{Assume } D = 2.0 \text{ m} \quad D = 2.0 \text{ m} \quad D = 2.0 \text{ m}$$

Then

$$V = 0.55 (2)^{0.64} = 0.55 (1.558) = 0.857 \text{ m/s}$$

$$A = \frac{Q}{V}$$

$$A = \frac{15}{0.857} = 17.50 \text{ m}^2$$

For trapezoidal section,

$$A = BD + z D^2$$

where

$$z = 0.65 \quad z = 0.65$$

$$D = 2 \text{ m} \quad D = 2 \text{ m}$$

$$17.5 = 2B + 0.65 (4) = 2B + 2.6B$$

$$B = 7.45 \text{ m}$$

$$P = B + 2D\sqrt{1+z^2} = 7.45 + 4\sqrt{1+0.65^2}$$

$$P = 7.45 + 4(1.193) = 7.45 + 4(1.193)$$

$$P = 12.22 \text{ m}$$

$$R = \frac{A}{P} = \frac{17.5}{12.22} = 1.43 \text{ m}$$

$$V = \frac{1}{n} R^{2/3} S^{1/2} = \frac{1}{0.023} (1.43)^{2/3} (0.0002)^{1/2} / 2 = 43.48 (1.27) (0.01414) = 43.48 (1.27) (0.01414)$$

$$V = 0.78 \text{ m/s}$$

$$Q = AV = 17.5 \times 0.78 = 13.65 \text{ m}^3/\text{s}$$

This is less than 15 m³/s, so revise the trial depth.

Trial 2:

$$\text{Take } D = 2.2 \text{ m}$$

Using the same procedure:

$$\text{Velocity (Kennedy): } 0.91 \text{ m/s}$$

$$A = \frac{15}{0.91} = 16.48 \text{ m}^2$$

$$16.48 = 2.2B + 0.65(2.2)^2$$

$$B \approx 6.06 \text{ m}$$

Checking with Manning gives:

$$R \approx 1.44 \text{ m} \quad R \approx 1.44 \text{ m} \quad R \approx 1.44 \text{ m}$$

$$V \approx 0.79 \text{ m/s} \quad V \approx 0.79 \text{ m/s} \quad V \approx 0.79 \text{ m/s}$$

The discharge is still slightly low; after a few iterations, the section converges.

9. a. What are the common building construction materials used for the construction of residential buildings? Describe one building material in details.
- b. Write short notes on (any two)
 - i. Water cement ratio of concrete mix
 - ii. Mixing concrete
 - iii. Bulking sand
 - iv. Curing of concrete

Answer:

The materials used in residential building construction should provide strength, durability, safety, economy, and comfort. The following are the most common building construction materials and their uses.

- i. **Cement:**

- Cement is the fine powder which sets after few hours when mixed with water and then hardens in a few days into a solid, strong material.
- It consists of 60-67% of lime, 17-25% of silica, 3-8% of alumina, 3-5% of calcium sulphate, 1-5% of iron oxide, 1-4% of magnesium oxide and 1-2% of alkali by weight.
- It is used in foundations, columns, beams, slabs, masonry work, plastering, and flooring.
- **Advantages:**
 - High compressive strength
 - Durable
 - Good binding property
 - Fire resistant

ii. Bricks:

- Bricks are the block of tampered clay molded to suitable shapes and sizes in its plastic stage dried in the sun and burnt if desired so as to make them strong, hard and durable.
- They are made by burning clay in kilns.
- Made up of 50 to 60% silica, 20 to 30% alumina or clay, 4 to 6% of lime, 4 to 6% of iron oxide and 1% of magnesia by weight of brick.
- **Uses:**
 - Load-bearing walls
 - Partition walls
 - Boundary walls
- **Advantages:**
 - Durable
 - Good thermal insulation
 - Fire resistant
 - Economical

iii. Stone:

- The rock quarried from the quarries is stone.
- The rock possessing suitable properties often find its use in building construction which is known as building stones.
- Natural stone is used where high strength and durability are required.
- Good building stone should absorb the water, preferably less than 5% and is discarded when absorbs the water more than 10%.
- **Uses:**
 - Foundations
 - Retaining walls
 - Masonry work
 - Flooring and cladding
- **Advantages:**
 - Very strong
 - Long service life
 - Weather resistant

iv. Concrete:

- Concrete is a heterogeneous mixture of binding medium and aggregate particles i.e. **cement, sand, coarse aggregate, and water.**
- Normally, the volume of concrete is occupied by constituents:

Cement= 70-75%

Water= 15-25%

Cement= 10—15%
- **Uses:**
 - Foundations

- o Columns
- o Beams
- o Slabs
- o Staircases
- **Advantages:**
 - o High compressive strength
 - o Durable
 - o Can be cast into any shape
 - o Low maintenance

v. Steel:

- Steel is an intermediate form between cast iron and wrought iron.
- It consists of upto 1.5% carbon content.
- Depending upon their carbon content steel is classified as:

Mild steel- if the carbon content is from 0.1-0.3%

Medium carbon steel- if the carbon content is from 0.3-0.5%.

High carbon steel- if the carbon content is between 0.5-1.5%.

- It is also used in roof trusses, frames, and grills.
- **Advantages:**
 - o High tensile strength
 - o Ductile
 - o Recyclable
 - o Easy to fabricate

vi. Wood:

- Timber is obtained from trees and is widely used in residential buildings.
- Generally, the raw wood contains 100-200% of moisture content which is brought to 10-20% by seasoning.
- **Uses:**
 - o Doors
 - o Windows
 - o Roof trusses
 - o Furniture
 - o Flooring
- **Advantages:**
 - o Easily available and has high salvage value.
 - o Good aesthetic appearance
 - o Good thermal insulation and shock absorbent
- **Disadvantages:**
 - o Vulnerable to termites
 - o Can decay if not properly treated

Short notes:

i. Water cement ratio of concrete mix:

- The water–cement ratio (W/C ratio) is the ratio of the weight of water to the weight of cement used in a concrete mix. Formula for the water content is given by:

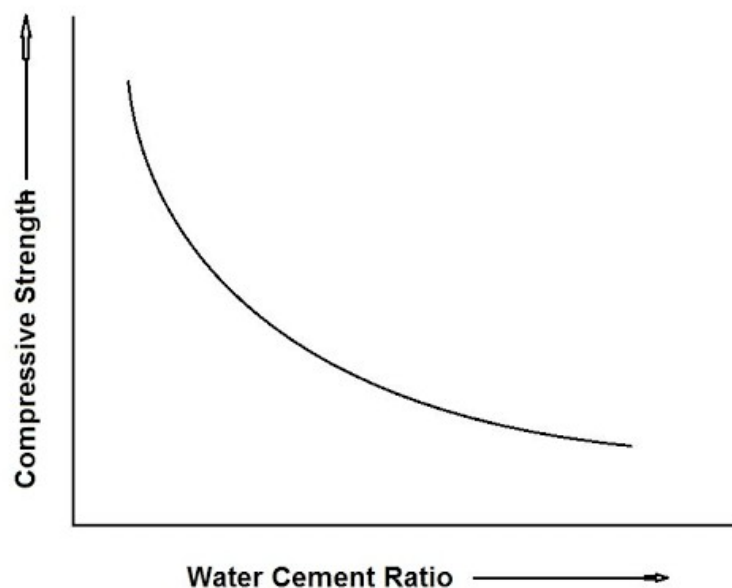
$$W / C \text{ Ratio} = \frac{\text{Weight of Water}}{\text{Weight of Cement}}$$

- Controls the strength, durability, and workability of concrete.

- In terms of volume, water used in litre per bag of cement (50kg) is taken as W/C ratio by volume.
- In terms of weight, quantity of water to be used per unit weight of cement is known as W/C ratio by weight.
- Effect of water-cement ratio:

Low W/C Ratio	High W/C Ratio
Higher strength of concrete	Lower strength of concrete
Better durability	Less durable
Lower workability	Higher workability
Less permeability	More permeability
Difficult compaction	Higher shrinkage and cracking

Type of Work	W/C Ratio
High-strength concrete	0.35–0.45
RCC work	0.40–0.55
PCC work	0.50–0.60



The compressive strength of concrete is inversely proportional to the water–cement ratio, provided proper compaction and curing are achieved.

ii. Mixing concrete:

Mixing of concrete is the process of uniformly combining cement, sand, coarse aggregate, water, and admixtures to obtain a homogeneous concrete mix.

Objectives:

- Uniform distribution of materials.
- Proper coating of aggregates with cement paste.
- Obtain required strength and workability.

Methods of mixing of concrete:

a. Hand mixing:

- Done on a watertight platform.
- Suitable for small works.

b. Machine mixing:

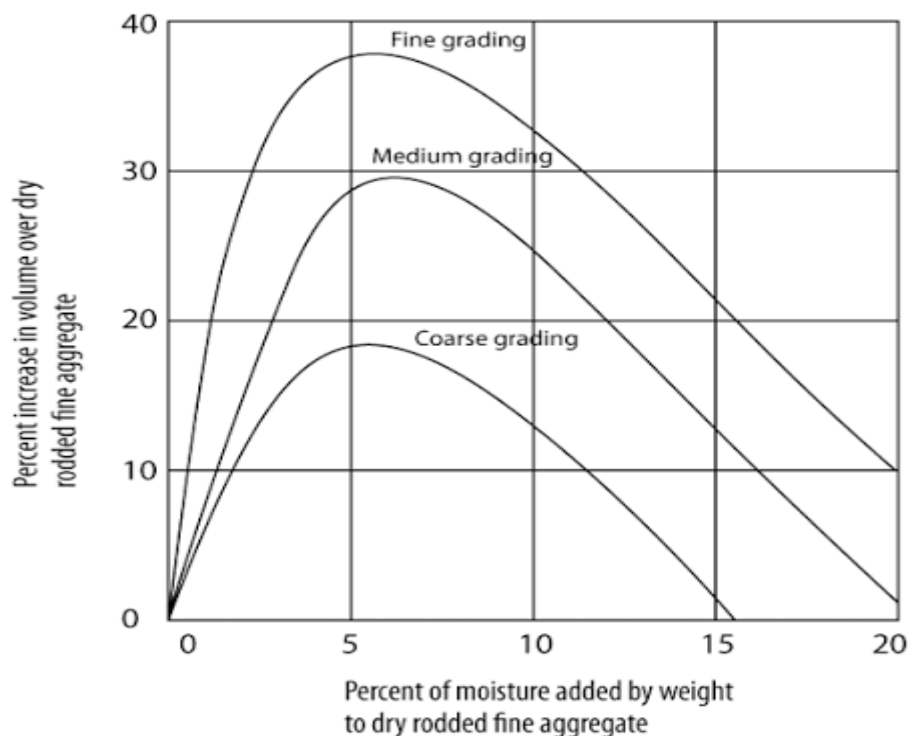
- Done using a concrete mixer.
- Used for medium and large projects.
- Produces better quality concrete.

Normally, mixing of the ingredients is done for 1.5–3 minutes after all materials are in the mixer.



iii. Bulking of sand:

- The increase in volume of sand due to the presence of the moisture content is called bulking of sand.
- Bulking of sand is caused by the films of water pushing the sand particles apart due to surface tension.
- Bulking phenomenon occurs much more in case of fine sands rather than in coarse ones.
- Maximum bulking occurs at **4–6% moisture content**.



- The phenomenon is important to consider when batching of aggregate is done by volume.
- Bulk sand occupies more volume than dry sand.
- To compensate the bulking effect extra sand is added. Otherwise, the concrete mix becomes deficient and will have tendency to segregate resulting in low yield of concrete.

iv. Curing of concrete:

- The process of preventing the loss of moisture from the concrete and maintaining the required temperature is known as curing of concrete.
- Curing strongly influences the strength, durability, water tightness, abrasion resistance, volume stability and resistance to freezing and thawing action.



- **Objectives:**
 - a. to prevent the loss of moisture from the concrete.
 - b. to maintain the favorable condition in temperature for a definite time.
- **Factors affecting the curing of concrete:**
 - a. Curing period:
 - b. Curing temperature:
 - c. Curing method and materials:
- **Methods of curing:**
 - a. Water curing:**
 - Ponding
 - Sprinkling
 - Wet covering (Hessian/Gunny bags)
 - b. Membrane curing:**
 - Plastic sheets
 - Curing compounds
 - c. Steam curing:** Used for precast concrete.
 - d. Heat curing:** Used in cold weather and precast industries.

10. As EIA National Guidelines now it is mandatory to conduct EIA for significant roads in Nepal. How does it help environmental as well as the road network? Give a critical analysis.

Answer:

- EIA (Environment Impact Assessment) is the most important tool for sustainable development approach. It is a detailed study or evaluation to be conducted to ascertain the potential adverse environmental impacts of a proposed project or activity and to identify measures to avoid, mitigate, or compensate for such impacts.
- EIA is done for:
 - a. Highways with the length of road greater than 50 km
 - b. Road tunnels with the length greater than 3 km.
 - c. Cable car ropeway longer than 5 km
 - d. Railway for public transportation.
 - e. Ropeway longer than 50 km.
- Includes:

Screening → Preparation of Scoping Document → Preparation and approval of work schedule and scoping document → Field study and public hearing → EIA Report Preparation → Recommendation and submission → final review and approval → Decision making
- EIA is required for large-scale development projects with significant environmental impacts that cannot be adequately assessed through a Brief Environmental Study (BES) or Initial Environmental Examination (IEE).

Benefits to the Road Network:

- **Impact Mitigation:** The primary goal of an EIA is to conduct a detailed study to ascertain if a proposal will have significant adverse impacts and to identify measures to avoid or mitigate them. This includes protecting natural, cultural, and social systems.
- **Biodiversity Conservation:** The process requires proponents to address potential threats to biological diversity, including eco-system, species, and genetic diversity, ensuring road paths do not irreparably harm Nepal's unique flora and fauna.
- **Pollution and Waste Management:** EIA standards regulate emissions (smoke, dust, gas) and pollution (noise, heat, radioactive rays) generated during construction and operation. It also mandates the proper management of hazardous substances and wastes.
- **Climate Change Resilience:** The Act integrates climate change management, requiring projects to consider adaptation and mitigation strategies to face challenges like gradual atmospheric changes and greenhouse gas emissions.

Benefits to the Road Network:

- **Optimal Planning through Alternative Analysis:** Proponents must perform a detailed analysis of alternatives regarding location, design, and technology. This ensures the chosen road alignment is the most appropriate and implementable, reducing the risk of project failure due to environmental or geological instability.
- **Project Sustainability:** By maintaining a balance between environment and development, the EIA process promotes the long-term viability of the road network, preventing environmental degradation that could lead to landslides or soil erosion affecting road integrity.
- **Structured Management Plans:** Every approved EIA must include an Environmental Management Plan (EMP). This plan outlines specific mitigation measures for both the construction and operational phases, ensuring the road remains functional and safe without causing unforeseen environmental harm.

While the mandatory EIA process provides a robust framework for sustainable development, it presents both strengths and systemic challenges:

- **Accountability and Public Oversight:** The requirement for public hearings ensures that local communities who are most affected have a voice in the project. Furthermore, the appointment of Environmental Inspectors to monitor compliance adds a layer of legal accountability.
- **Stringent Penalties:** The Act provides significant deterrents for non-compliance, including fines up to five million rupees for executing a project without an approved EIA or in a manner inconsistent with the approved report.

- **Administrative Complexity:** A potential drawback is the lengthy approval and inquiry process involving multiple committees and subject experts, which can lead to delays in critical infrastructure delivery.
- **Reliance on Consultant Quality:** The Act acknowledges risks regarding the "standards and quality" of reports. If a consultant prepares a substandard report, they can be barred for up to five years, but the underlying road project may already have suffered from poor initial planning.
- **Rigidity in Revisions:** If even minor changes are made to the physical infrastructure or design after approval, a Supplementary EIA is required, which may increase the administrative and financial burden on road development.